

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 919+840

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

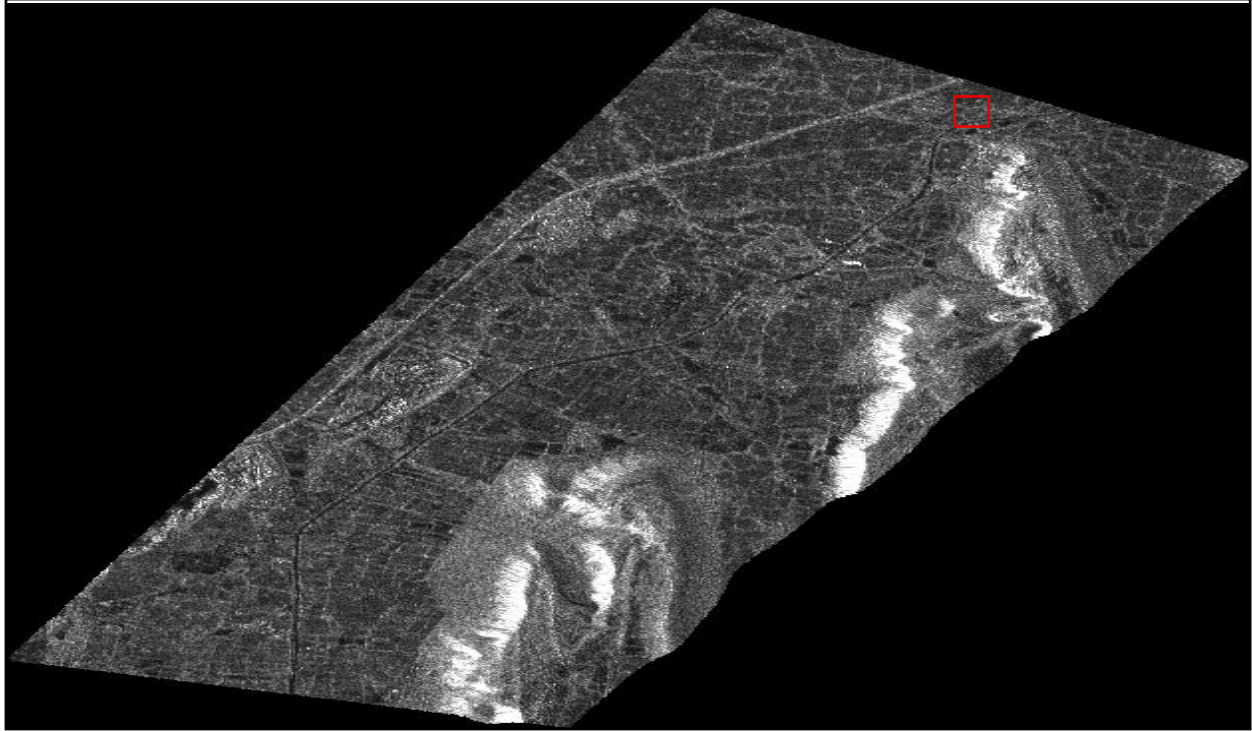
Flight Direction - DESCENDING

Polarization – VV+VH

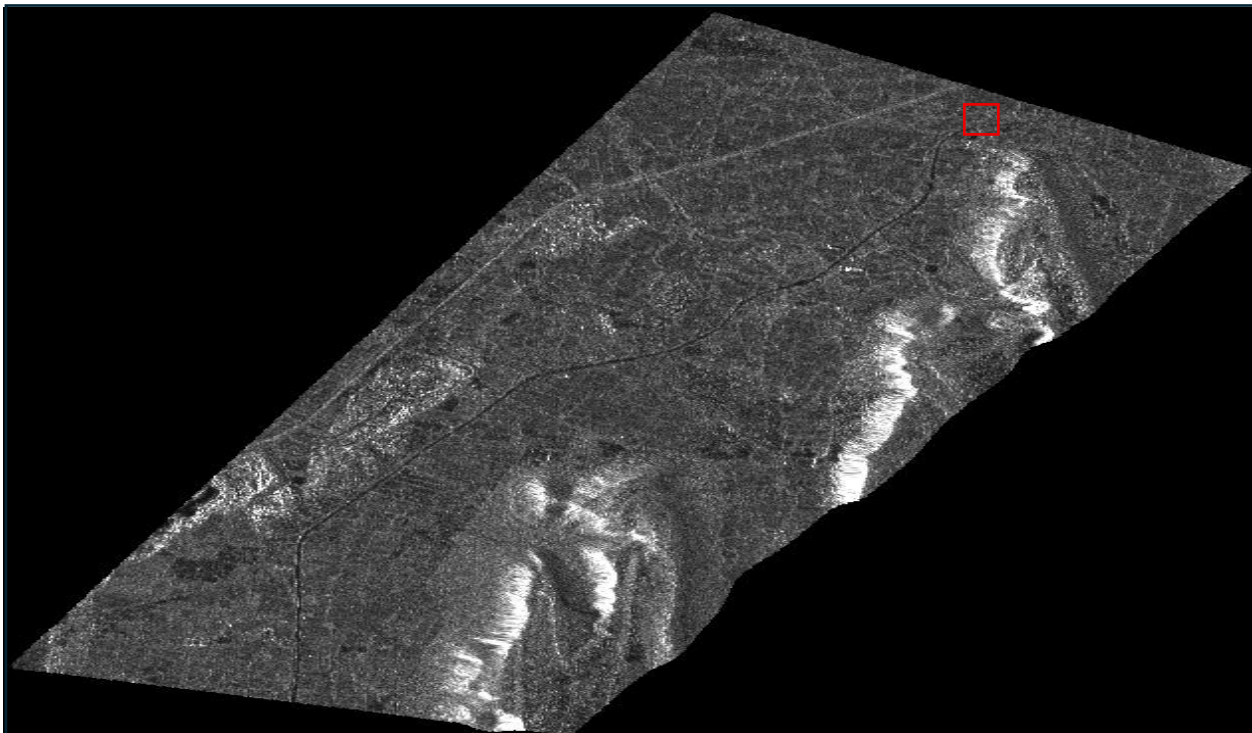
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

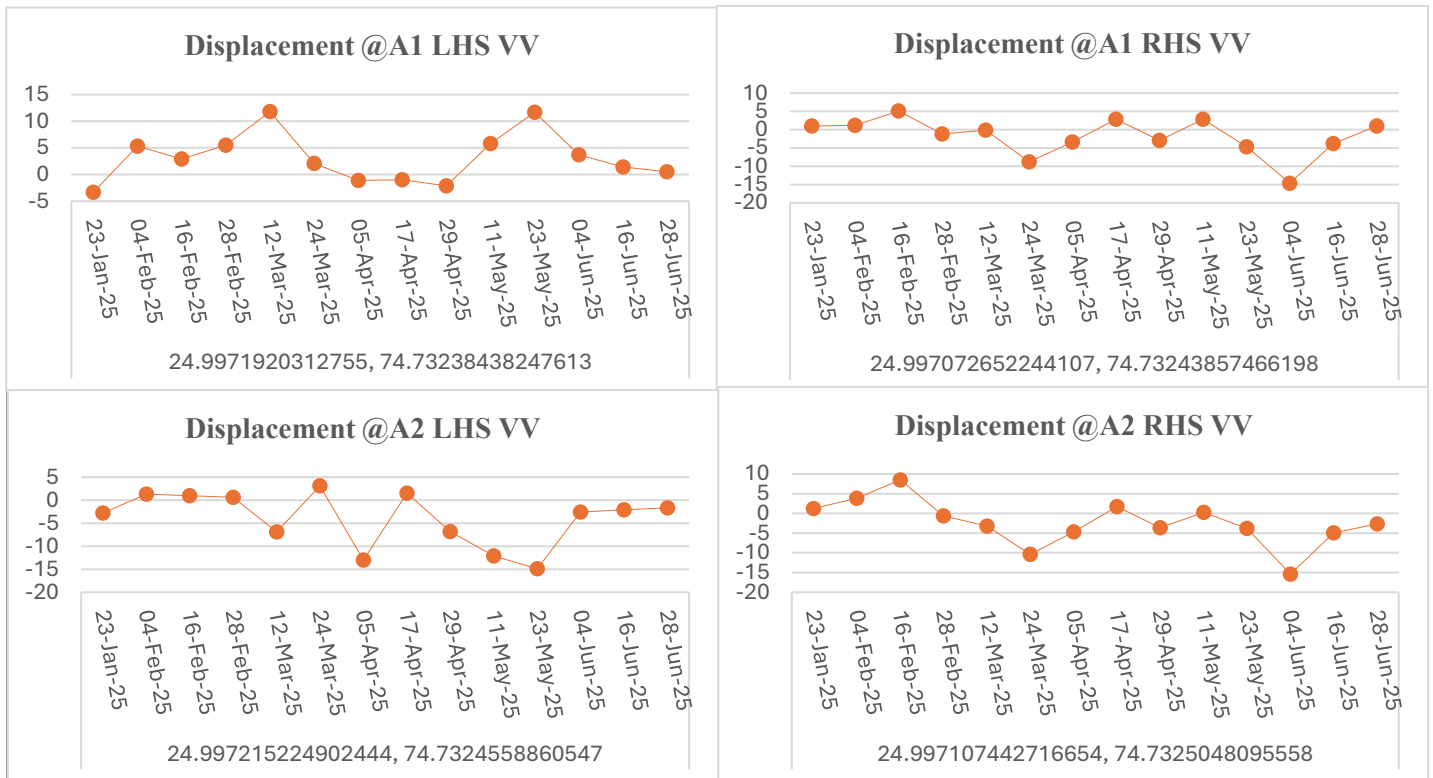
VUP – CH-917+840

Length – 6.62M

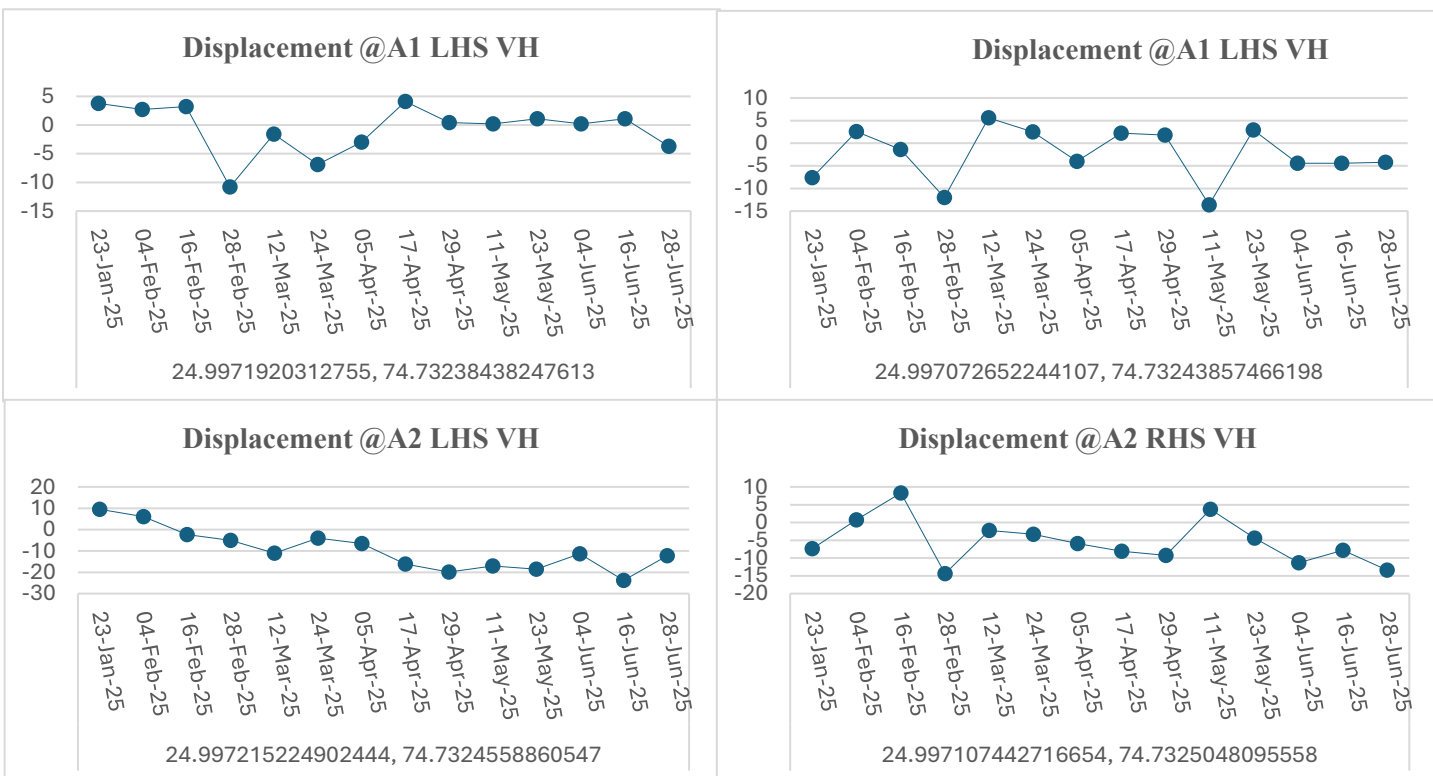


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

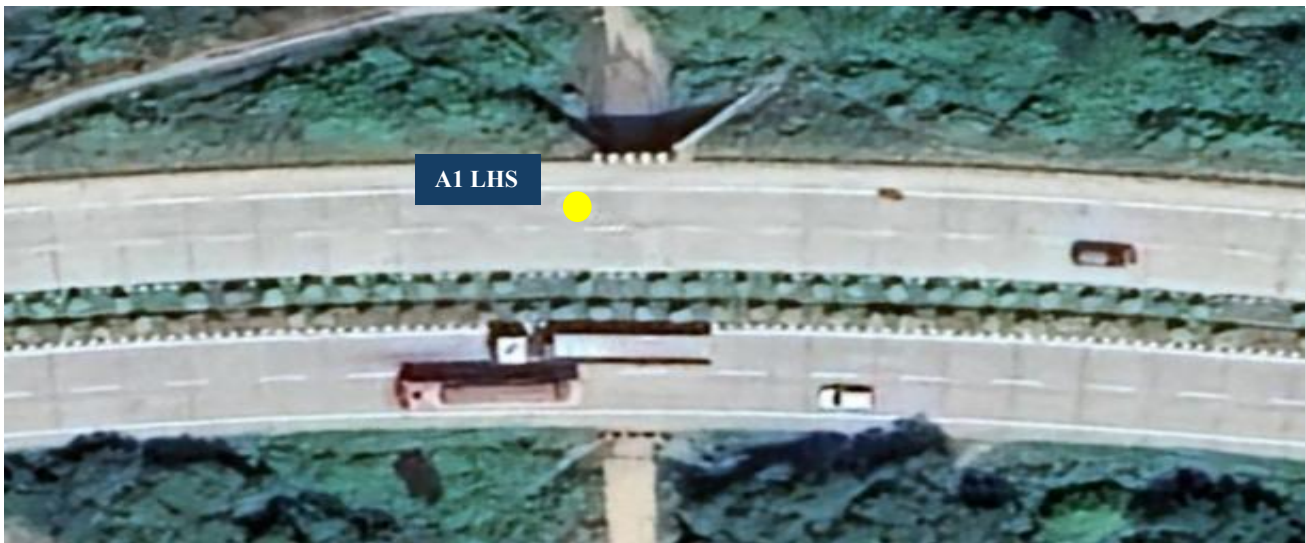
3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



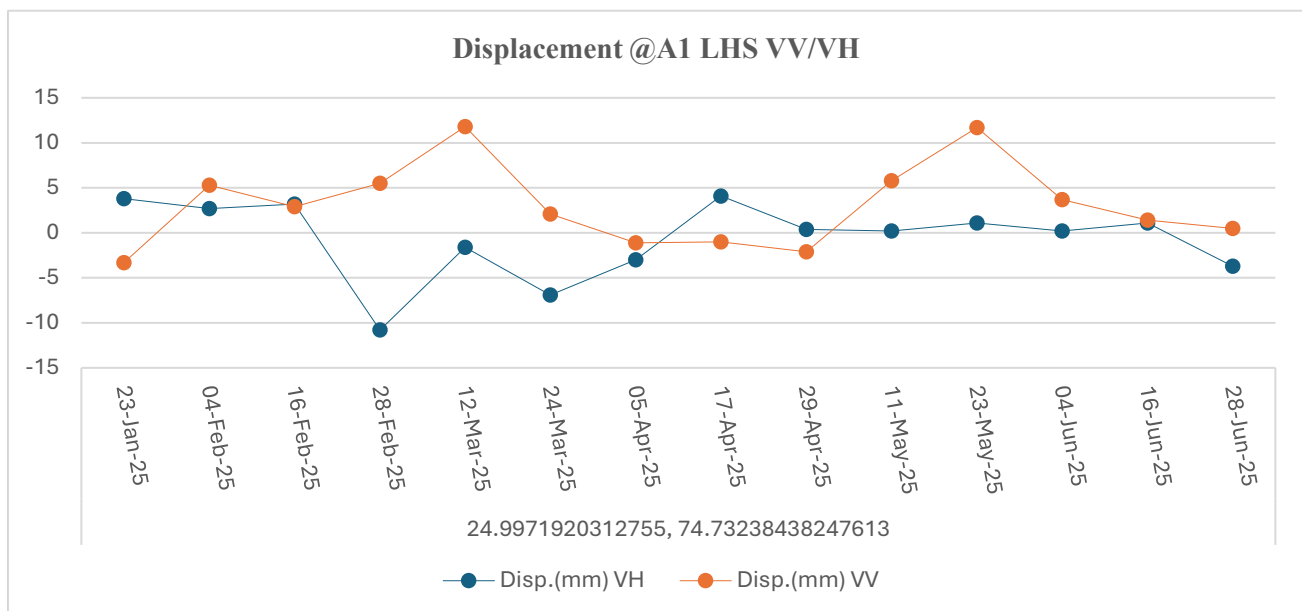
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.9971920312755, 74.73238438247613)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement ranges approximately from -4 mm to $+12$ mm.
- **Horizontal Displacement (VH)** Horizontal displacement shows moderate variability, generally ranging from approximately -11 mm to $+4$ mm.

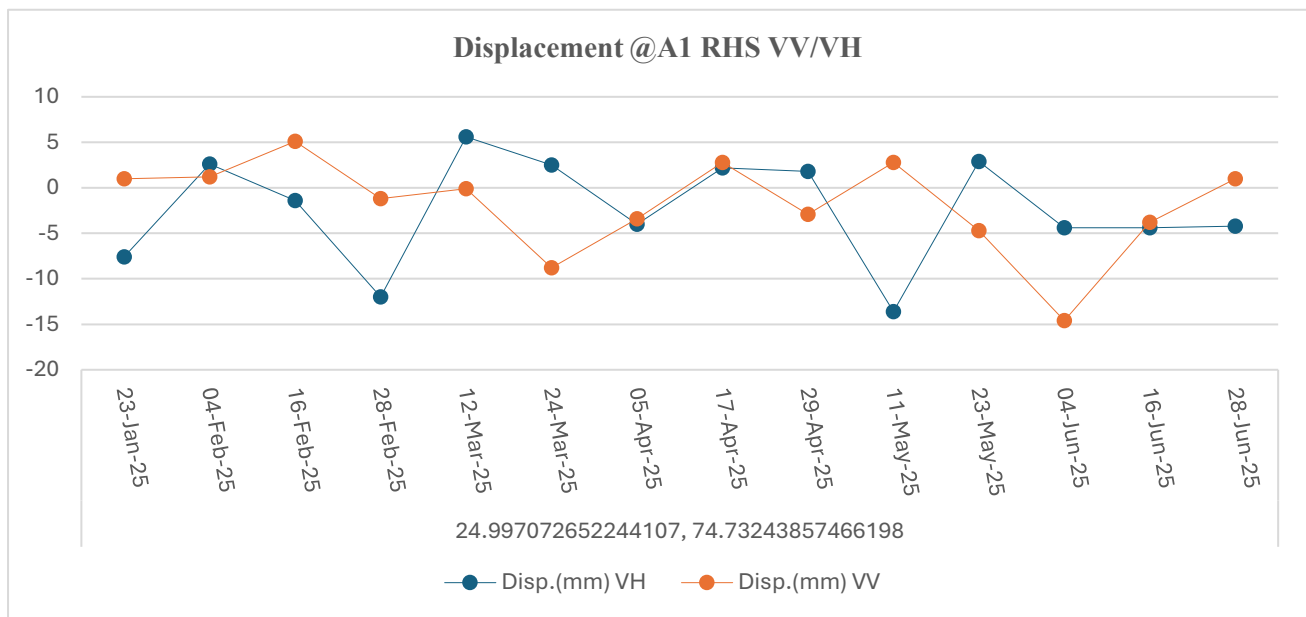
Critical Events & Anomalies:

- **February 2025:** A pronounced negative VH peak (around -11 mm) coincides with a moderate positive VV, indicating temporary lateral movement accompanied by vertical uplift.
- **May 2025:** VV exhibits another uplift peak while VH stabilizes close to zero, indicating localized vertical response without significant lateral involvement.
- **June 2025:** Both VH and VV converge toward small magnitudes, suggesting stabilization of movements and absence of progressive deformation.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.997072652244107, 74.73243857466198)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits noticeable variability, ranging approximately from -14 mm to +5 mm.
- **Vertical displacement (VV)** Vertical displacement ranges approximately from -15 mm to +5 mm, showing moderate vertical movement with alternating uplift and settlement phases over the monitoring period.

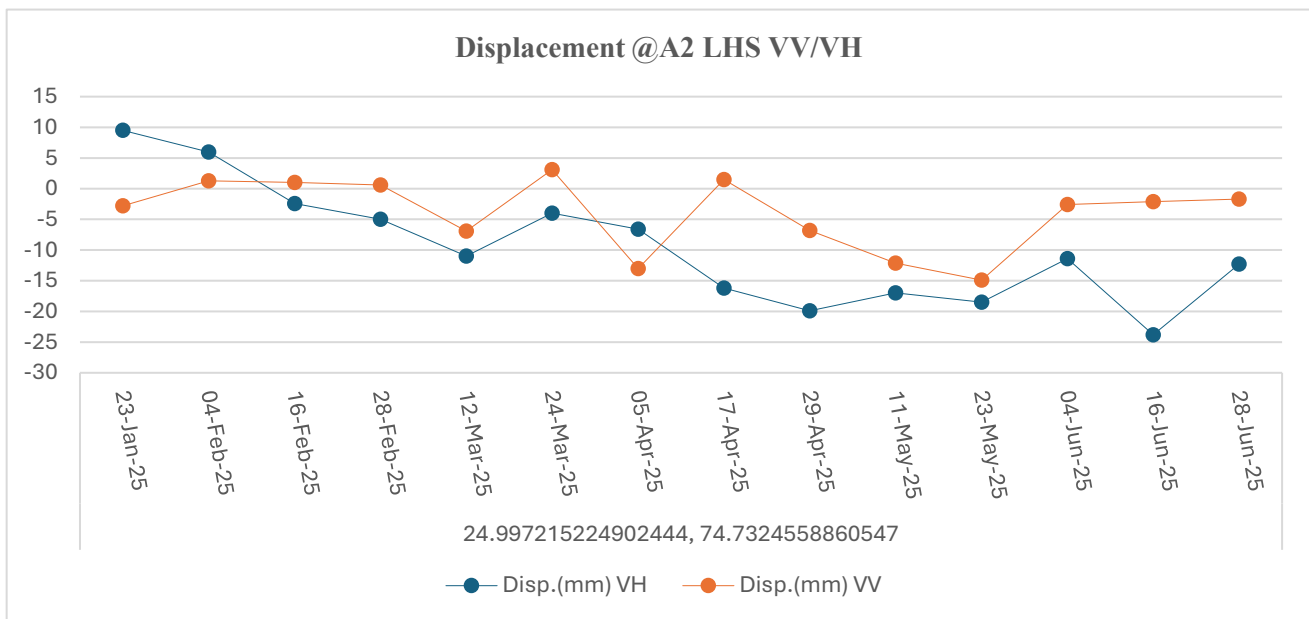
Critical Events & Anomalies:

- **February 2025:** A sharp negative VH dip occurs while VV remains near neutral to slightly positive, suggesting a localized lateral disturbance without significant vertical coupling.
- **March 2025:** VH reaches a local positive peak (around +5 mm) concurrent with VV transitioning downward, indicating a shift from uplift to settlement-dominated behaviour.
- **June 2025:** Both VH and VV stabilize near small negative to neutral values, indicating reduced displacement activity and overall stabilization of the structure.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.997215224902444, 74.7324558860547)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows a strong and progressive negative trend, transitioning from positive values to significant negative magnitudes.
- **Vertical Displacement (VV)** at A2 LHS exhibits alternating uplift and settlement phases, but with a clear dominance of negative values across most of the monitoring period.

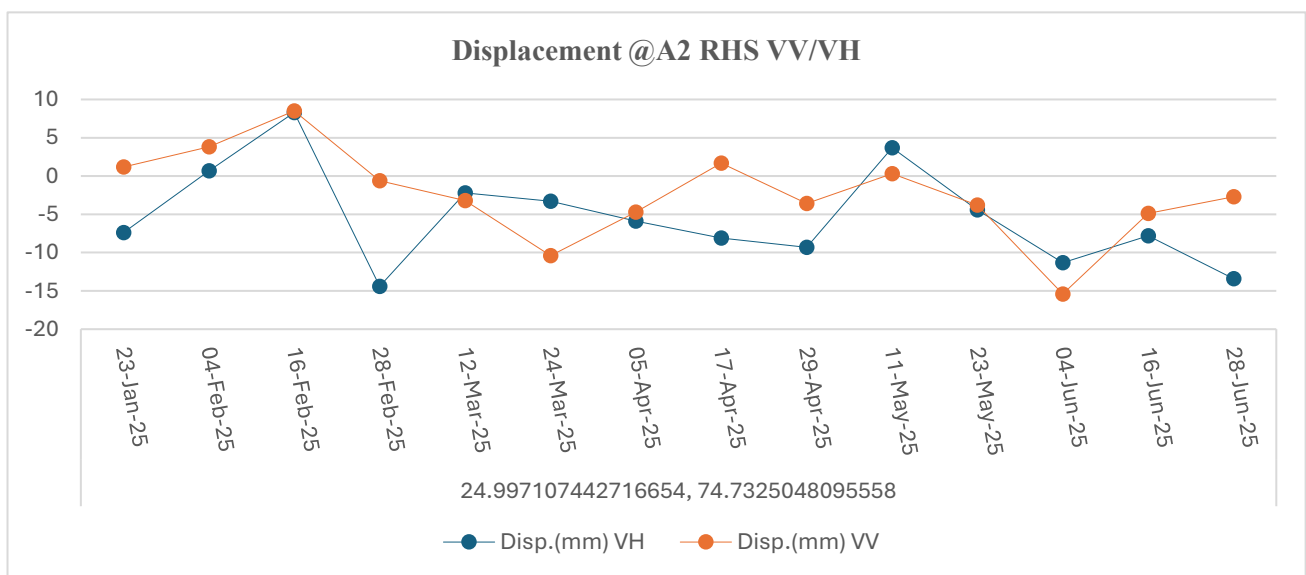
Critical Events & Anomalies:

- **January 2025:** VH starts at a positive peak, while VV is slightly negative, indicating initial lateral movement opposite to later trends, with minimal vertical settlement.
- **June 2025:** VH partially recovers to -12 mm, but VV stays negative, confirming no full structural recovery by the end of monitoring.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.997107442716654, 74.7325048095558)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement at A2 RHS exhibits alternating uplift and settlement behaviour, with values ranging approximately between +8 mm uplift and -15 mm settlement.
- **Horizontal (VH) displacement** Horizontal displacement shows moderate to high variability, ranging from approximately +8 mm to -15 mm.

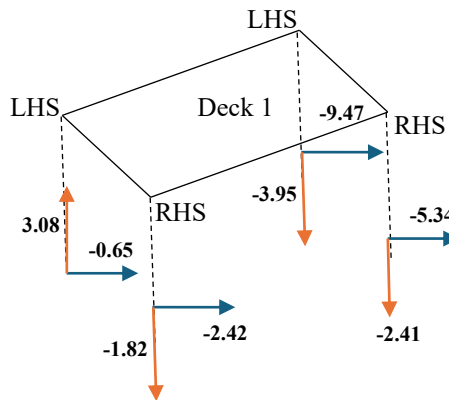
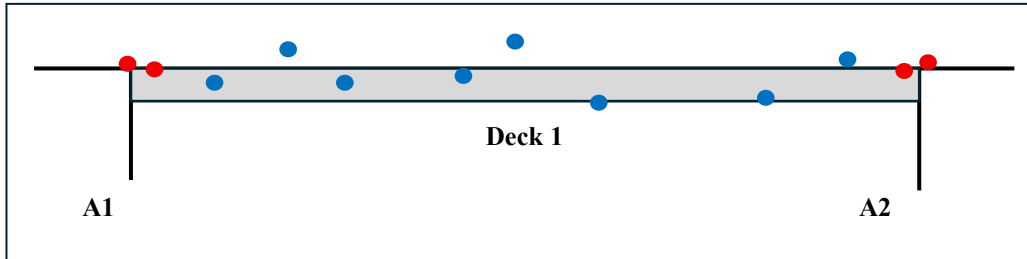
Critical Events & Anomalies:

- **January 2025:** VH starts in the negative range, while VV is slightly positive (+1 mm), indicating initial lateral offset with minimal vertical disturbance.
- **May 2025:** VH shows a short recovery to +3 mm, while VV approaches 0 mm, marking a brief stabilization window.
- **June 2025:** Both VH and VV remain negative, confirming no full recovery by the end of the monitoring period.

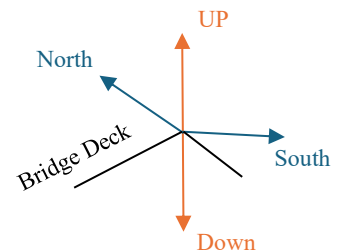
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab shows low to moderate vertical displacements, with both uplift and settlement observed across LHS and RHS supports. At the deck level (Deck 1), a downward displacement of approximately -3.95 mm is observed near the RHS, indicating minor localized settlement at slab level.

Horizontal Displacement Behaviour:

- Horizontal displacements are present but relatively restrained compared to earlier monitoring epochs. At the deck level near RHS, a horizontal displacement of approximately -9.47 mm is observed, indicating lateral movement predominantly toward the southward direction. The RHS mid-lower region exhibits a horizontal displacement of approximately -5.34 mm, confirming a consistent southward lateral bias on the RHS.

CONCLUSION:

- The bridge deck slab exhibits stable vertical behaviour, with all recorded vertical displacements remaining relatively small and within acceptable limits. Horizontal displacements are modest and directionally consistent, with slightly higher magnitudes on the RHS compared to the LHS. Overall structural response appears stable, with current displacement patterns indicative of normal operational behaviour rather than structural distress.